

TEST PLAN DOCUMENT

Project: VWO Login Application

URL: <https://app.vwo.com/#/login>

Prepared By: Senior QA Architect (20+ Years Experience)

Version: 1.0

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1. Application Understanding

The VWO login module enables users to authenticate using multiple mechanisms including email/password, Google authentication, SSO, and passkey-based login. The system supports password recovery via email and is expected to provide secure and reliable access control.

2. Executive Summary

This test plan defines the strategy to validate the VWO login system from functional, security, and performance perspectives. The goal is to ensure a stable, secure, and scalable authentication experience aligned with enterprise SaaS standards.

3. Objectives and Scope

Objectives

- Validate correctness of login workflows
- Ensure security against common vulnerabilities
- Assess performance under load

Scope

In Scope

- Login page UI
- Authentication flows
- Forgot password
- Multi-authentication mechanisms

Out of Scope

- Post-login dashboard
 - Backend APIs
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4. Assumptions and Dependencies

Assumptions

- Test accounts are available
- Authentication services are operational

Dependencies

- Google authentication service
 - SSO Identity Provider
 - Network stability
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5. Test Strategy and Test Design

Functional Testing

- Login success and failure scenarios
- Input validation

Security Testing

- Injection attacks
- Authentication validation

Performance Testing

- Response time validation
- Load and stress testing

Test Design Techniques

- Boundary value analysis
 - Negative testing
 - Equivalence partitioning
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6. RICEPOT Strategy

Role

Senior QA Automation Architect

Instructions

- Generate enterprise-grade test coverage
- Follow RTCFR format
- Avoid assumptions

Context

- Multi-authentication login system

Example

Standard structured test case format

Parameters

- High reliability

- Automation-ready

Output

- Test Plan
- Test Cases

Tone

Enterprise-grade QA documentation

7. Product Risk Analysis

Risk	Impact	Mitigation
Authentication failure	High	Extensive functional testing
Security vulnerabilities	Critical	Security testing suite
Third-party login issues	High	Mock testing
Performance degradation	High	Load testing

8. Prioritization and Environment

Prioritization

- High: Login success, security validation
- Medium: UI validations
- Low: Non-critical UI

Environment

- Browsers: Chrome, Firefox, Edge
 - OS: Windows, macOS
 - Tools: Playwright, JMeter/k6
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9. Test Data

- Valid user credentials
 - Invalid credentials
 - Edge case inputs
 - Security payload inputs
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10. Security and Accessibility

Security

- Input validation
- Authentication control
- Secure communication

Accessibility

- Keyboard navigation
 - Screen reader compatibility
 - Proper labeling
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11. Exploratory and Automation Strategy

Exploratory Testing

- Ad-hoc validation of edge cases
- UI behavior validation

Automation Strategy

- Playwright with Pytest
 - Page Object Model
 - CI/CD integration
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12. Entry / Exit / Suspension / Resumption Criteria

Entry Criteria

- Application accessible
- Test data available

Exit Criteria

- All high-priority defects resolved
- Test cases executed

Suspension Criteria

- Critical defect blocking testing

Resumption Criteria

- Blocking issues resolved
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13. Defect Management

- Tool: JIRA or equivalent
 - Severity classification: Critical, High, Medium, Low
 - Lifecycle: Open → In Progress → Fixed → Verified → Closed
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14. Deliverables and Responsibilities

Deliverables

- Test Plan
- Test Cases
- Execution Report
- Defect Report

Responsibilities

- QA Team: Test execution
 - Dev Team: Defect fixes
 - QA Lead: Approval
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15. Metrics

- Test execution rate
 - Defect density
 - Pass/Fail ratio
 - Defect leakage
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16. Coverage Gaps and Clarifications

- Session behavior Not specified
 - Error message handling Not specified
 - Performance thresholds Not specified
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17. Test Case Generation Handoff

Test cases are created in RTCFR format and handed over to:

- Automation team for scripting
 - QA team for execution
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18. References

- VWO official documentation
 - Authentication best practices
 - OWASP security guidelines
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End of Document